

3. Perform the following tasks using git-bash

- Initialize a Git repository for a data science project
- Push it to GitHub to enable version control
- Create a new branch named GIT-EXAM-1 and create new file named as program2 and write a program to check if given string is palindrome
- Create a python file name as program1.py and write a python program to print squares of numbers from 1 to 10
- Commit the code and push the changes and check the changes on remote repository(GitHub)
- Create a new branch named GIT-EXAM-2
- Pull the changes from branch GIT-EXAM-1
- create new file named as program2.py and write a program to check if given string is palindrome
- Modify the code in program1.py(to demonstrate conflicts)
- Commit the changes and push them and demonstrate the difference of changes in master/main, GIT-EXAM-1 and GIT-EXAM-2 branch
- Create a pull request to merge the changes from GIT-EXAM-2 branch to GITEXAM-1 branch

a. Open **Git Bash** and run:

```
mkdir data-science-project
```

```
cd data-science-project
```

```
git init
```

✅ b. Push it to GitHub

- Go to GitHub and create a new repository (e.g., data-science-project)
- Connect local repo to GitHub:

```
git remote add origin https://github.com/your-username/data-science-project.git
```

OR

```
git clone https://github.com/your-username/data-science-project.git
```

```
cd data-science-project
```

Open bash in cloned repo

```
echo "# Data Science Project" > README.md
```

```
git add .
```

```
git commit -m "Initial commit"
```

```
git branch -M main
```

```
git push -u origin main
```

✅ Step 3: Create Branch GIT-EXAM-1

```
git checkout -b GIT-EXAM-1
```

✅ Step 4: Create program2.py (Palindrome)

```
touch program2.py  
code .
```

```
# Palindrome check  
string = input("Enter string: ")  
if string == string[::-1]:  
    print("Palindrome")  
else:  
    print("Not Palindrome")
```

✅ Step 5: Create program1.py (Squares 1–10)

```
touch program1.py  
code .
```

```
# Squares from 1 to 10  
for i in range(1, 11):  
    print(i*i)
```

✅ Step 6: Commit & Push (GIT-EXAM-1)

```
git add .  
git commit -m "Added palindrome and squares programs"  
git push origin GIT-EXAM-1
```

✅ Step 7: Create Branch GIT-EXAM-2

```
git checkout -b GIT-EXAM-2
```

✅ Step 8: Pull Changes from GIT-EXAM-1

```
git pull origin GIT-EXAM-1
```

✅ Step 9: Modify program2.py (Palindrome - different logic)

```
code .      👉 Updated code:
```

```
# Palindrome (method 2)  
string = input("Enter string: ")  
rev = "".join(reversed(string))
```

```
if string == rev:  
    print("Palindrome")  
else:  
    print("Not Palindrome")
```

✅ Step 10: Modify program1.py (to create conflict)

code . 🖱 Modified code:

```
# Modified squares program
for i in range(1, 11):
    print("Square of", i, "is", i*i)
```

✅ Step 11: Commit & Push (GIT-EXAM-2)

```
git add .
git commit -m "Modified program1 and program2"
git push origin GIT-EXAM-2
```

✅ Step 12: Show Differences

```
git diff main GIT-EXAM-1
git diff GIT-EXAM-1 GIT-EXAM-2
```

✅ Step 13: Create Pull Request (IMPORTANT)

1. Open repo , Click **Compare & Pull Request**(git exam 2)
2. Base → GIT-EXAM-1
3. Compare → GIT-EXAM-2
4. Click **Create Pull Request** → **Merge**

🎯 Final Output

- ✓ main → basic repo
- ✓ GIT-EXAM-1 → original programs
- ✓ GIT-EXAM-2 → modified programs